

Shyamoli Textile Engineering College
Department of Computer Science and Engineering
CSE-4237: Digital Image Processing

Lec-11: Color Image Processing

Digital Image Processing by Rafael C. Gonzalez and Richard Eugene Woods (Chapter-6)

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2.5 Color Model

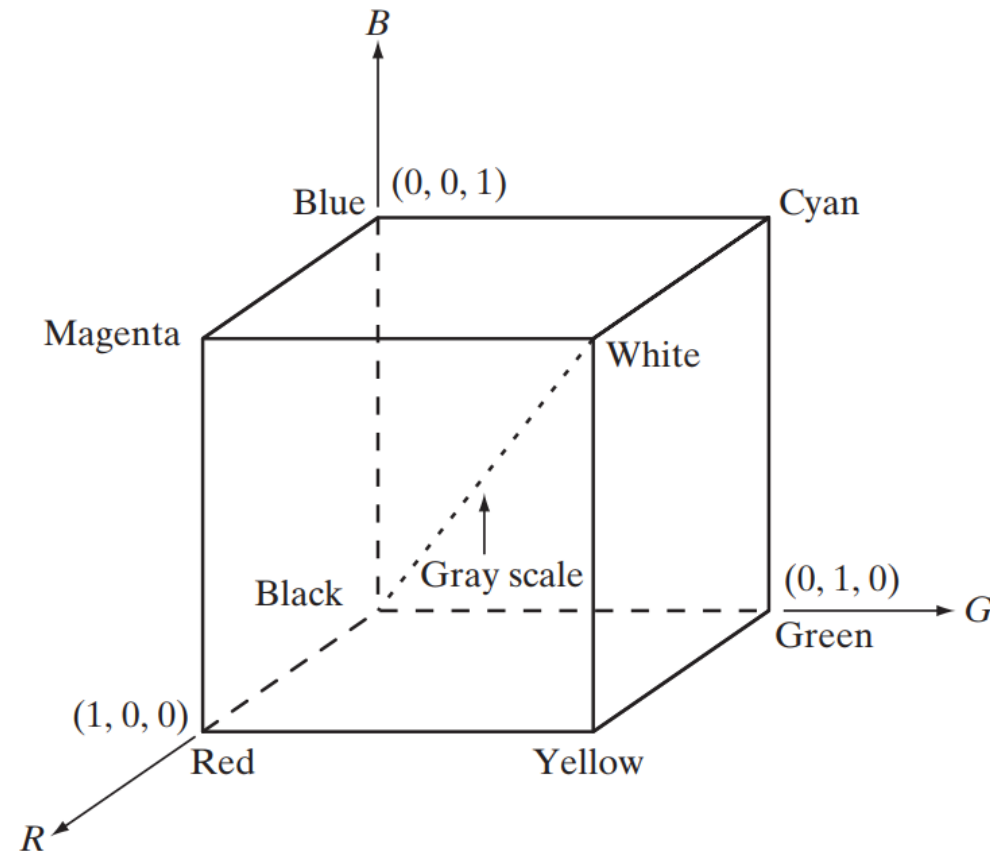
A color model is a system used to describe a color.

The hardware-oriented models most commonly used in practice are the **RGB** (red, green, blue) model for color monitors and a broad class of color video cameras; the **CMY** (cyan, magenta, yellow) and **CMYK** (cyan, magenta, yellow, black) models for color printing; and the **HSI** (hue, saturation, intensity) model, which corresponds closely with the way humans describe and interpret color.

The **HSI model** also has the advantage that it decouples the color and gray-scale information in an image, making it suitable for many of the gray-scale techniques

2.5.1 RGB Model

RGB is an additive color model. In the RGB model, each color appears in its primary spectral components of red, green, and blue. This model is based on a Cartesian coordinate system.



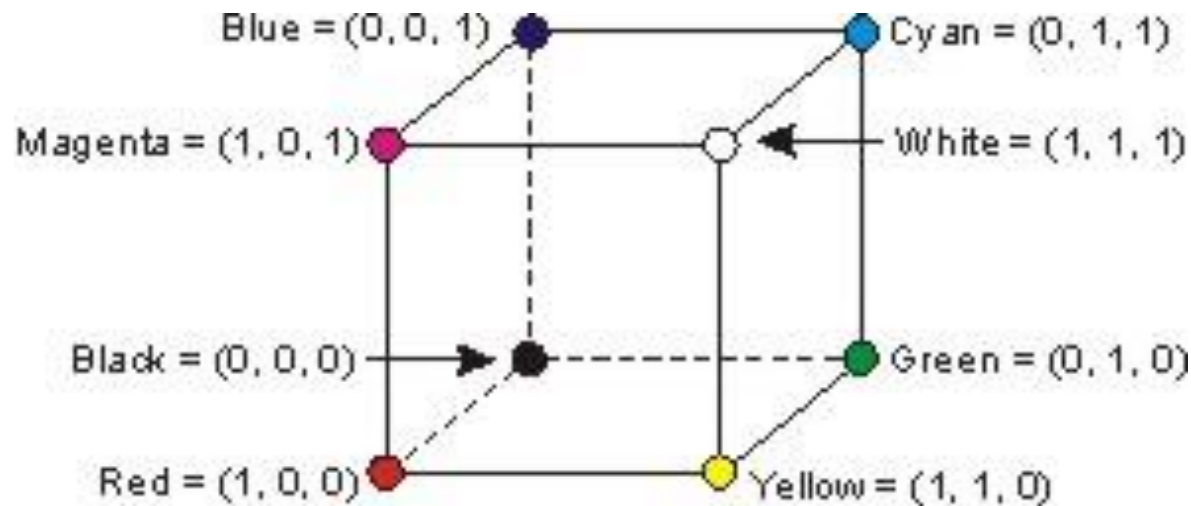
The color subspace of interest is the cube in which:

RGB primary values are at three corners;

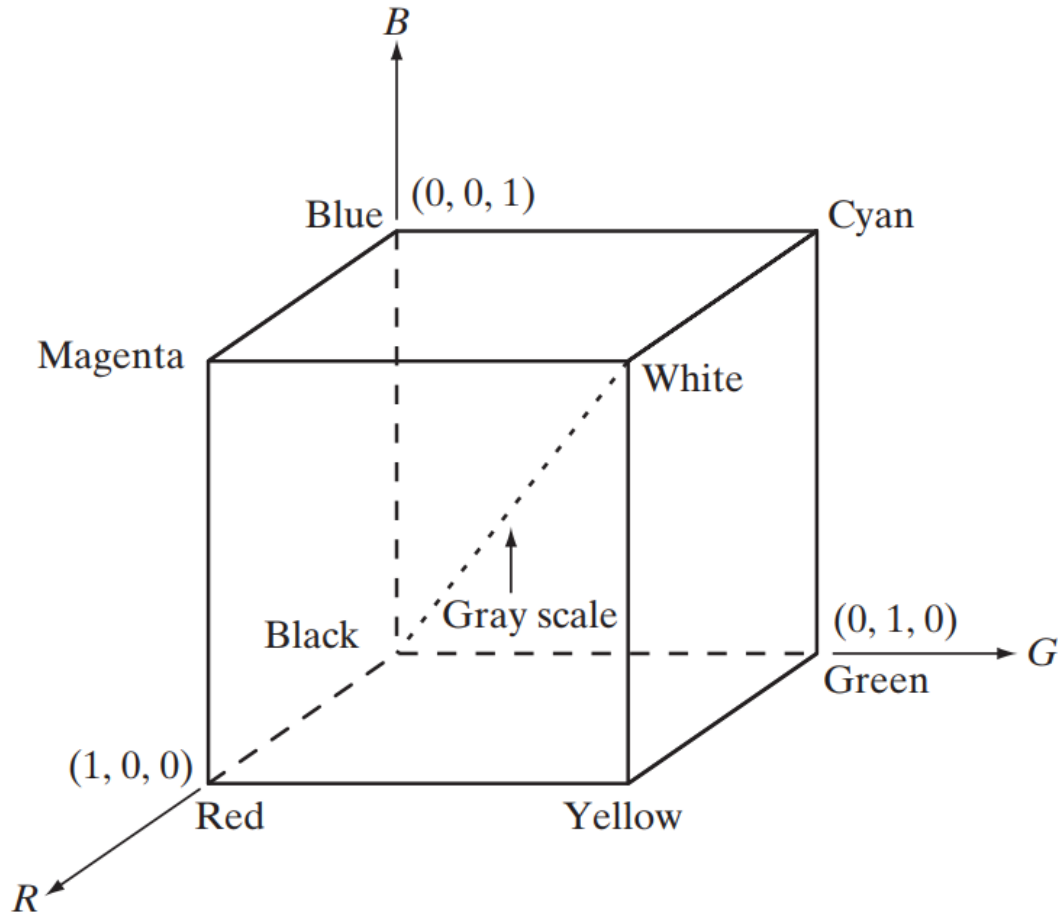
the *secondary colors cyan, magenta, and yellow are at three other corners;*

black is at the origin; and

white is at the corner farthest from the origin.



In this model, the gray scale (points of equal RGB values) extends from black to white along the line joining these two points. For convenience, the assumption is that all color values have been normalized so that the cube is the unit cube. That is, all values of R, G, and B are assumed to be in the range $[0, 1]$.



The number of bits used to represent each pixel in RGB space is called the **pixel depth**.

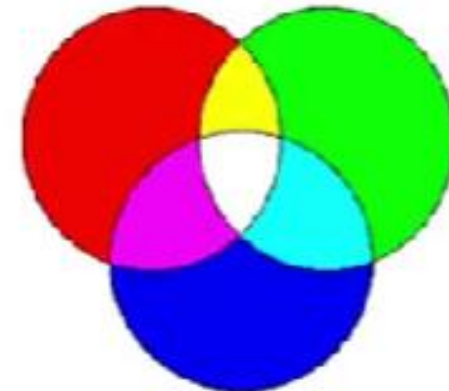
Consider an RGB image in which each of the red, green, and blue images is an 8-bit image. Under these conditions each RGB color pixel [that is, a triplet of values (R, G, B)] is said to have a depth of 24 bits (3 image planes times the number of bits per plane).

The term full-color image is used often to denote a 24-bit RGB color image.

The total number of colors in a 24-bit RGB image is

$$(2^8)^3 = 16,777,216$$

24 bit = 8 bit + 8 bit + 8 bit
red green blue



The *cube is a solid, composed* of the $(2^8)^3 = 16,777,216$ colors mentioned in the preceding paragraph.



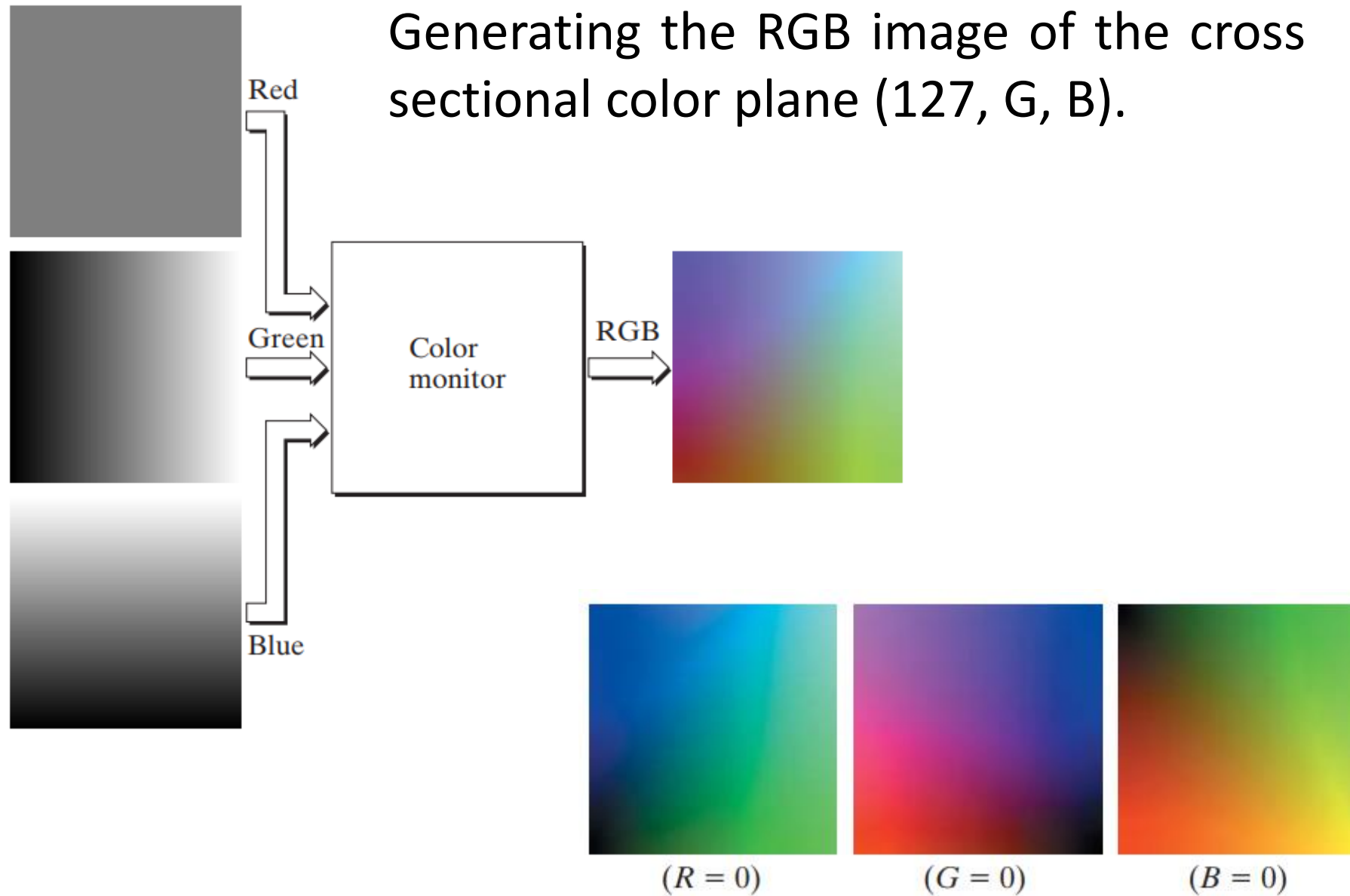
A convenient way to view these colors is to *generate color planes* (faces or cross sections of the cube).

This is accomplished simply by *fixing one of the three colors* and *allowing the other two to vary*.

For instance, a cross-sectional plane through the center of the cube and parallel to the GB-plane is the plane $(127, G, B)$ for $G, B = 0, 1, 2, \dots, 255$.



Here *we used the actual pixel values* rather than the mathematically convenient normalized values in the range $[0, 1]$ because the former values are the ones actually *used in a computer to generate colors*.



The three hidden surface planes in the color cube



Tools



Brushes

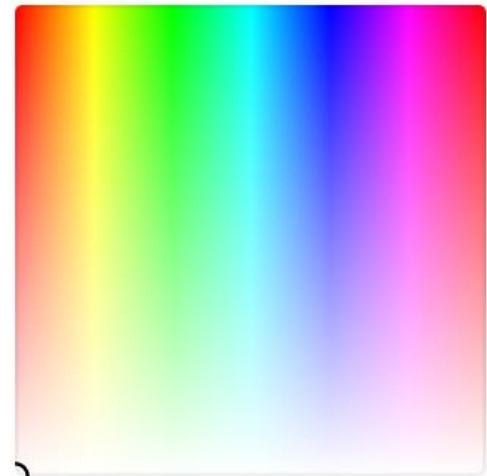


Image Creator



Layers

Edit colors



#000000

RGB

0

Red

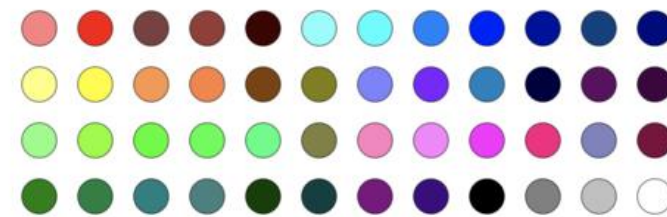
0

Green

0

Blue

Basic colors

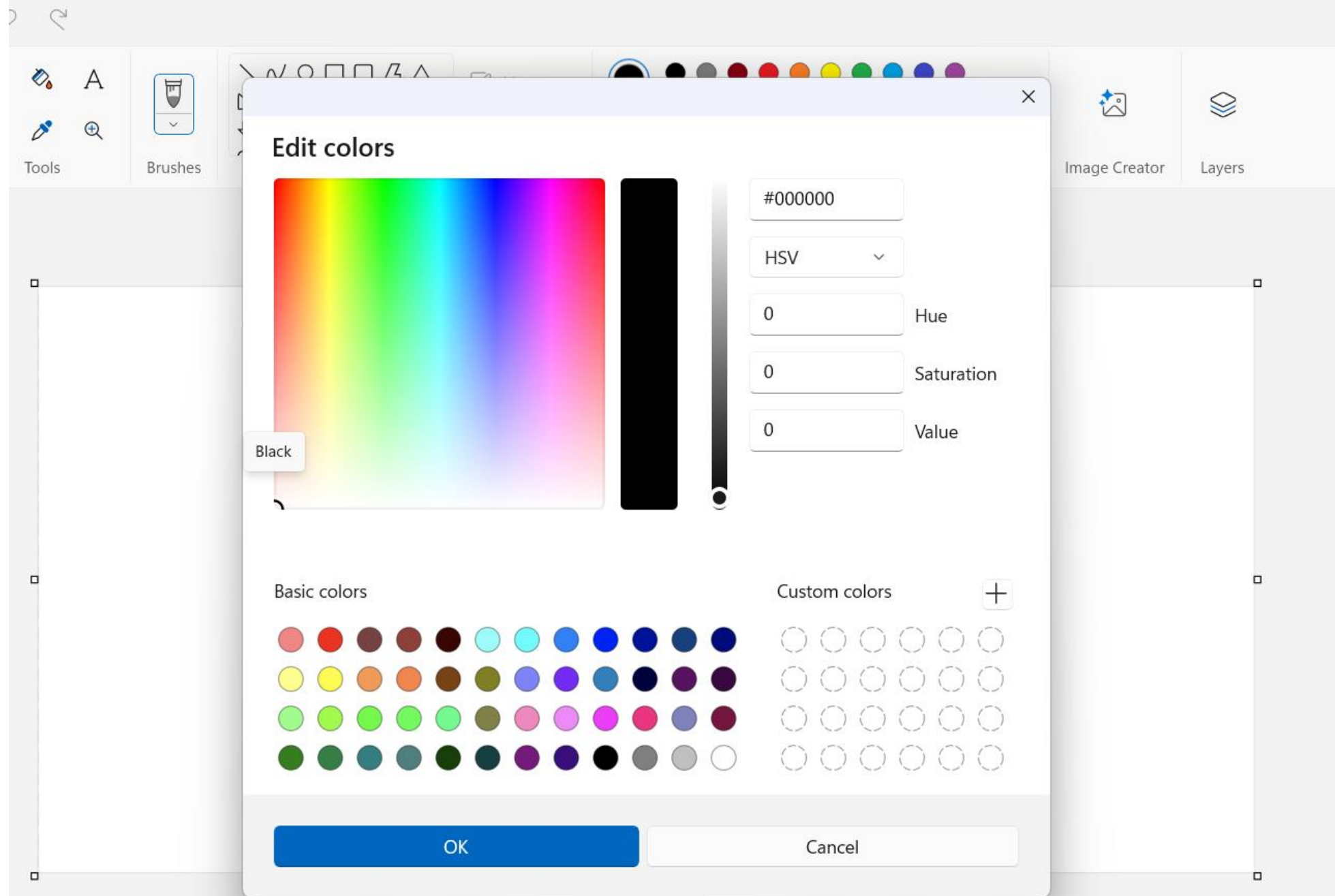


Custom colors



OK

Cancel



2.5.2 The CMY and CMYK Color Models

CMY / CMYK are subtractive color model.

CMY stands for Cyan, Magenta, and Yellow, the three primary colors used in subtractive color mixing.

Each color in the CMY model absorbs (subtracts) one primary color of light:

- ☐ Cyan absorbs red light.
- ☐ Magenta absorbs green light.
- ☐ Yellow absorbs blue light.

In theory, *mixing all three (Cyan, Magenta, and Yellow) should produce black, as all light is absorbed.*

Most devices that deposit colored pigments on paper, such as color printers and copiers, require CMY data input or perform an RGB to CMY conversion internally.

Cyan = White – Red

Magenta = White – Green

Yellow = White – Blue

This conversion is performed using the simple operation

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

Where, again, the assumption is that all color values have been normalized to the range $[0, 1]$.

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

Equation demonstrates that light reflected from a surface coated with pure cyan does not contain red (that is, in the equation). Similarly, pure magenta does not reflect green, and pure yellow does not reflect blue.

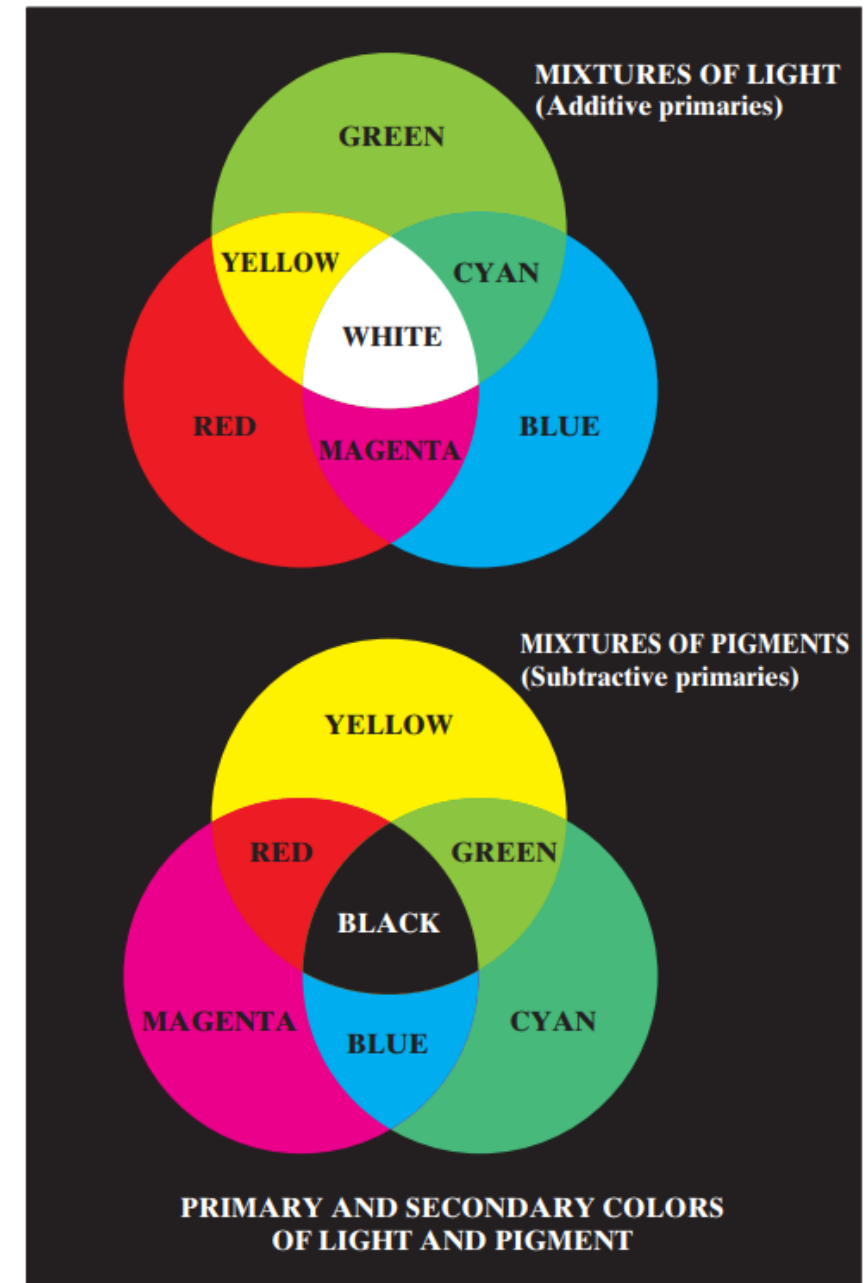
*** Equation also reveals that RGB values can be obtained easily from a set of CMY values by subtracting the individual CMY values from 1.

❖ As indicated earlier, in image processing this color model is used in connection with generating hardcopy output, so the inverse operation from CMY to RGB generally is of little practical interest.

According to Fig, equal amounts of the pigment primaries, cyan, magenta, and yellow should produce black. In practice, combining these colors for printing produces a muddy-looking black.

So, in order to produce true black (which is the predominant color in printing), a fourth color, black, is added, giving rise to the CMYK color model.

Thus, when publishers talk about “four-color printing,” they are referring to the three colors of the CMY color model plus black.



2.5.3 The HSI Color Model

As we have seen, creating colors in the RGB and CMY models and changing from one model to the other is a **straightforward process**. As noted earlier, these color systems are ideally suited for hardware implementations.

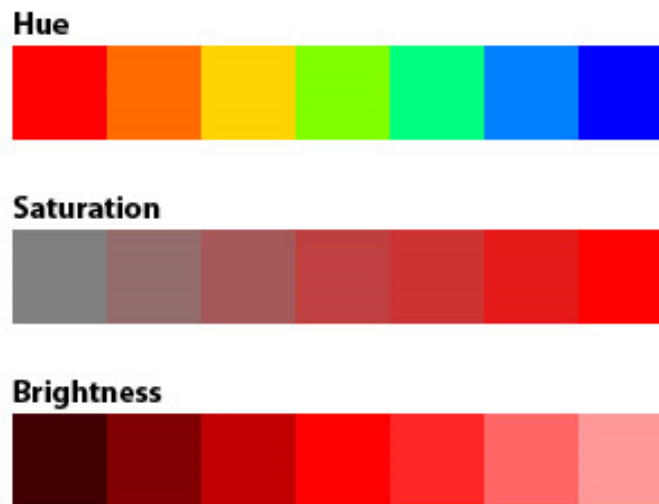
In addition, the RGB system matches nicely with the fact that the human eye is strongly perceptive to red, green, and blue primaries.

Unfortunately, the RGB, CMY, and other similar color models are not well suited for describing colors in terms that are practical for human interpretation.

When humans view a color object, we describe it by its hue, saturation, and brightness.

Hue is a color attribute that describes a pure color (pure yellow, orange, or red), whereas **saturation** gives a measure of the degree to which a pure color is diluted by white light.

Brightness is a subjective descriptor that is practically impossible to measure. It embodies the achromatic notion of intensity and is one of the key factors in describing color sensation.



Why HSI model?

We do know that intensity (gray level) is a most useful descriptor of monochromatic images. This quantity definitely is measurable and easily interpretable.

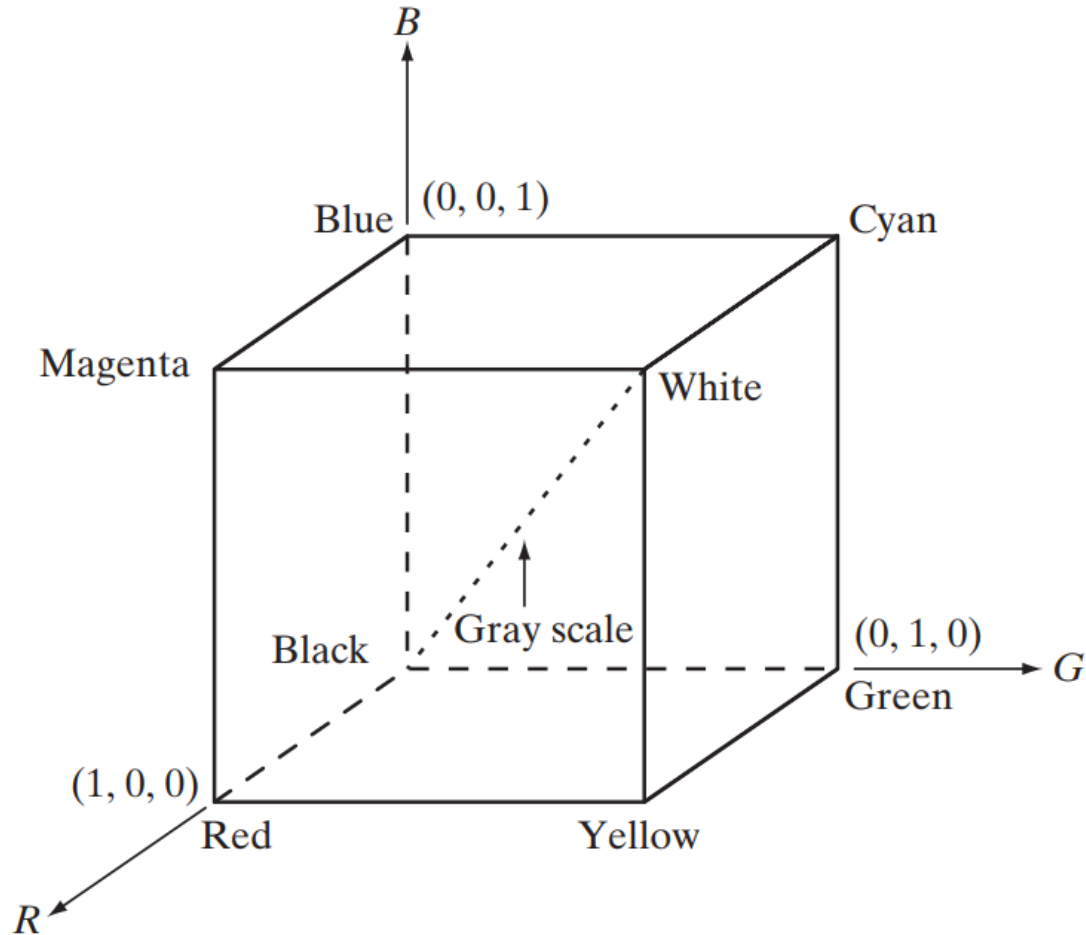
The model we are about to present, called the HSI (hue, saturation, intensity) color model, decouples the intensity component from the color-carrying information (hue and saturation) in a color image.

As a result, the HSI model is an ideal tool for developing image processing algorithms based on color descriptions that are natural and intuitive to humans, who, after all, are the developers and users of these algorithms.

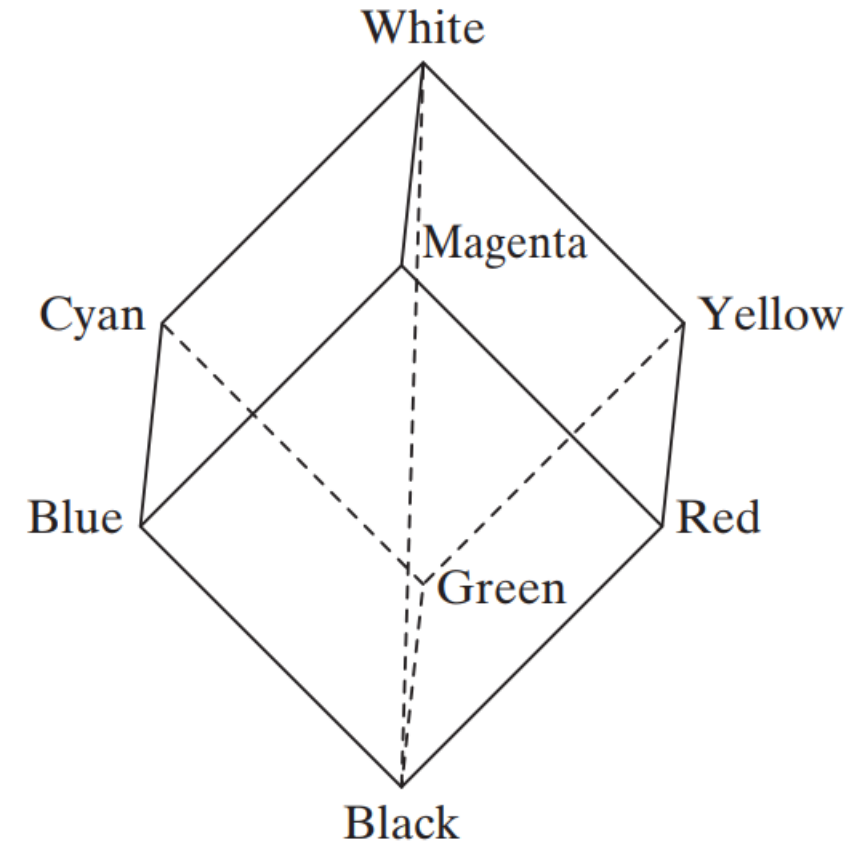
We can summarize by saying that RGB is ideal for image color generation (as in image capture by a color camera or image display in a monitor screen), but its use for color description is much more limited. The material that follows provides an effective way to do this.

An RGB color image can be viewed as three monochrome intensity images (representing red, green, and blue), so it should come as no surprise that we should be able to extract intensity from an RGB image.

This becomes rather clear if we take the color cube and stand it on the black (0, 0, 0) vertex, with the white vertex (1, 1, 1) directly above it

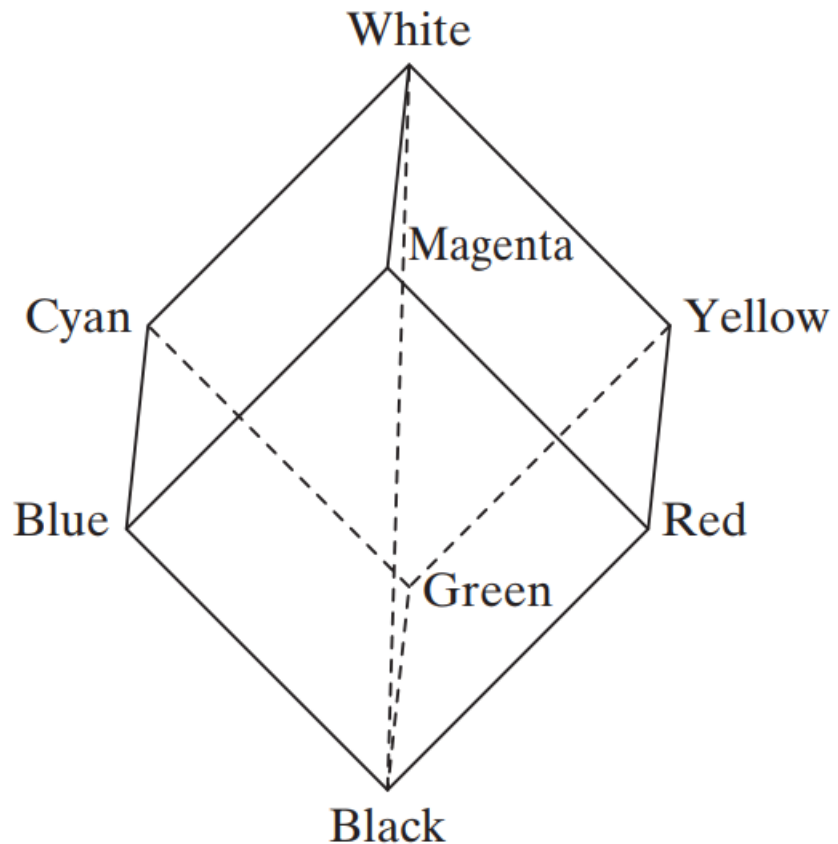


The intensity (gray scale) is along the line joining these two vertices



The line (intensity axis) joining the black and white vertices is vertical

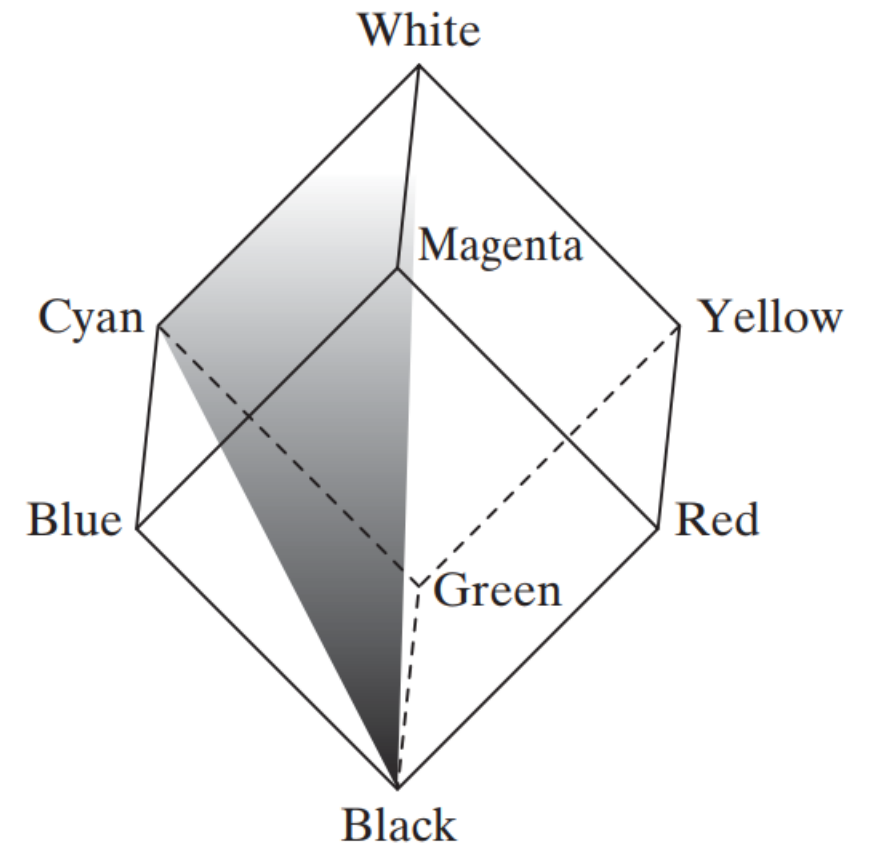
Thus, if we wanted to determine the intensity component of any color point in Fig, we would simply pass a plane perpendicular to the intensity axis and containing the color point. The intersection of the plane with the intensity axis would give us a point with intensity value in the range $[0, 1]$.



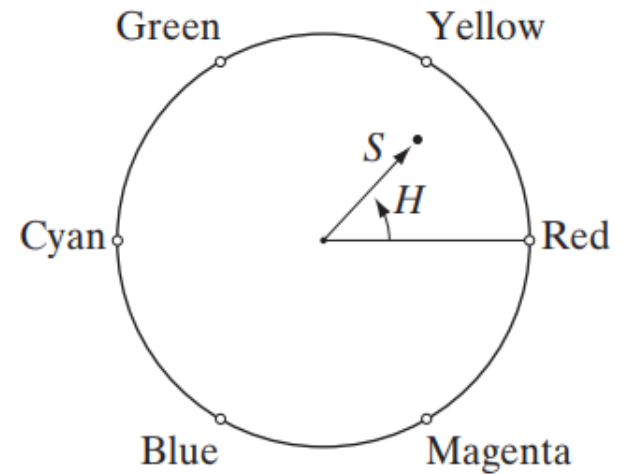
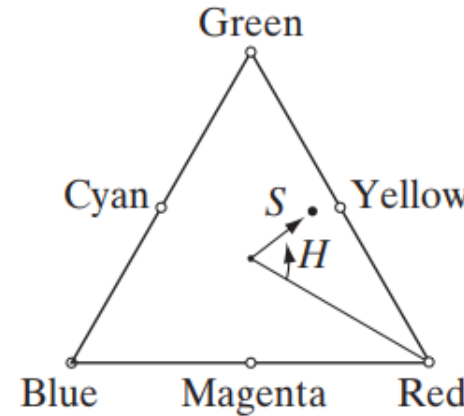
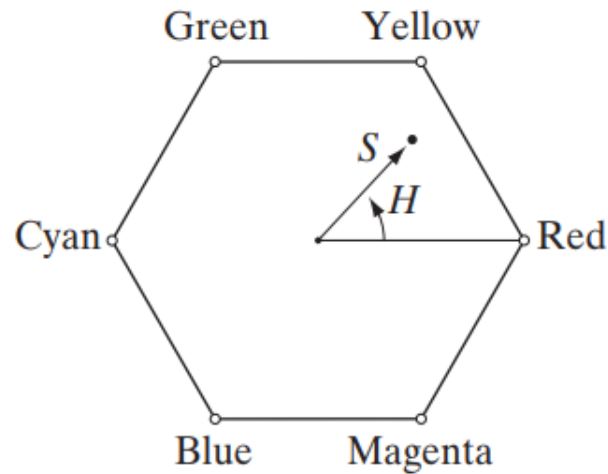
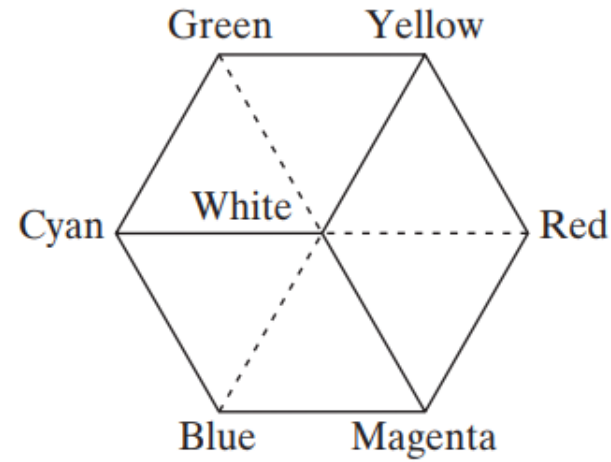
We also note with a little thought that the saturation (purity) of a color increases as a function of distance from the intensity axis. In fact, the saturation of points on the intensity axis is zero, as evidenced by the fact that all points along this axis are gray.

In order to see how hue can be determined also from a given RGB point, consider Fig, which shows a plane defined by three points (black, white, and cyan). The fact that the black and white points are contained in the plane tells us that the intensity axis also is contained in the plane.

Furthermore, we see that all points contained in the plane segment defined by the intensity axis and the boundaries of the cube have the same hue (cyan in this case).

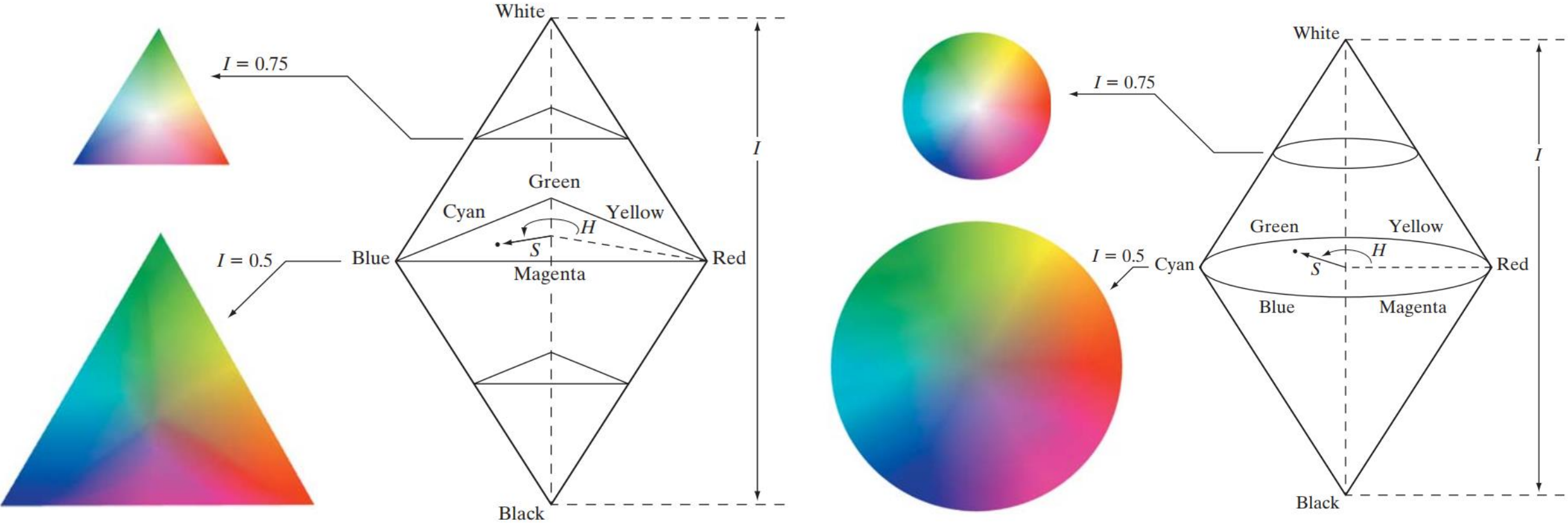


In this plane we see that the primary colors are separated by 120° . The secondary colors are 60° from the primaries, which means that the angle between secondaries also is 120° .



Hue and saturation in the HSI color model. The dot is an arbitrary color point. The angle from the red axis gives the hue, and the length of the vector is the saturation.

Figure shows the HSI model based on color triangles and also on circles.



Converting colors from RGB to HSI

Given an image in RGB color format, the H component of each RGB pixel is obtained using the equation

$$H = \begin{cases} \theta & \text{if } B \leq G \\ 360 - \theta & \text{if } B > G \end{cases}$$

$$\theta = \cos^{-1} \left\{ \frac{\frac{1}{2}[(R - G) + (R - B)]}{[(R - G)^2 + (R - B)(G - B)]^{1/2}} \right\}$$

The saturation component is given by

$$S = 1 - \frac{3}{(R + G + B)} [\min(R, G, B)]$$

Finally, the intensity component is given by

$$I = \frac{1}{3}(R + G + B)$$